

Savitzky-Golay split command-lag search

rotor median publish period [s]: 0.00499773025513
gimbal median publish period [s]: 0.00499868392944
initial lags [s]: {'rotor': 0.004997730255126953, 'gimbal': 0.004998683929443359}
smooth half-width multipliers: [4.0, 2.0, 1.0, 0.5]

smooth width=4: rotor=0.2s gimbal=0.0321992446s cost=1.49733540171 lag_gradient={'rotor':
-0.8555150290780236, 'gimbal': -2.256867716571298e-06} FD=[{'direction':
'negative_gradient', 'scheme': 'forward', 'step': 1.3877787807867988e-17,
'analytic_l2_norm': 7.529392537932048, 'finite_difference_l2_norm': 0.0,
'difference_l2_norm': 7.529392537932048, 'relative_error': 1.0}, {'direction':
'rotor_delay_seconds', 'scheme': 'forward', 'step': 1.3877787807814457e-17,
'analytic_l2_norm': 7.529393301188546, 'finite_difference_l2_norm': 0.0,
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'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 15.86438729705662, 'finite_difference_l2_norm': 15.864387194029318,
'difference_l2_norm': 6.047398739801596e-07, 'relative_error': 3.8119333741452424e-08}]
smooth width=2: rotor=0.2s gimbal=0.0321264937s cost=1.49892605742 lag_gradient={'rotor':
-0.8725465059269738, 'gimbal': 3.194760922198281e-06} FD=[{'direction': 'negative_gradient',
'scheme': 'forward', 'step': 1.387778780790988e-17, 'analytic_l2_norm': 8.278759109305707,
'finite_difference_l2_norm': 0.0, 'difference_l2_norm': 8.278759109305707, 'relative_error':
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'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 17.379165593118394, 'finite_difference_l2_norm': 17.379164188849657,
'difference_l2_norm': 4.466770636596212e-06, 'relative_error': 2.570187051077362e-07}]
smooth width=1: rotor=0.2s gimbal=0.0316364562s cost=1.49891270051 lag_gradient={'rotor':
-0.6227882702640668, 'gimbal': -5.016068825747233e-06} FD=[{'direction':
'negative_gradient', 'scheme': 'forward', 'step': 1.3877787808269676e-17,
'analytic_l2_norm': 10.27566798200938, 'finite_difference_l2_norm': 0.0,
'difference_l2_norm': 10.27566798200938, 'relative_error': 1.0}, {'direction':
'rotor_delay_seconds', 'scheme': 'forward', 'step': 1.3877787807814457e-17,
'analytic_l2_norm': 10.275670420866682, 'finite_difference_l2_norm': 0.0,
'difference_l2_norm': 10.275670420866682, 'relative_error': 1.0}, {'direction':
'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 20.25865321204618, 'finite_difference_l2_norm': 20.25864376604837,
'difference_l2_norm': 2.8790053734260613e-05, 'relative_error': 1.4211237752538011e-06}]
smooth width=0.5: rotor=0.2s gimbal=0.030154945s cost=1.49843732636 lag_gradient={'rotor':
-0.3675320533801921, 'gimbal': 1.1522156223270352e-05} FD=[{'direction':
'negative_gradient', 'scheme': 'forward', 'step': 1.387778781468469e-17, 'analytic_l2_norm':
14.00432986018053, 'finite_difference_l2_norm': 0.0, 'difference_l2_norm':
14.00432986018053, 'relative_error': 1.0}, {'direction': 'rotor_delay_seconds', 'scheme':
'forward', 'step': 1.3877787807814457e-17, 'analytic_l2_norm': 14.004339069373975,
'finite_difference_l2_norm': 0.0, 'difference_l2_norm': 14.004339069373975,
'relative_error': 1.0}, {'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step':
6.0554544523933395e-06, 'analytic_l2_norm': 18.227671917760137, 'finite_difference_l2_norm':
18.227646127616413, 'difference_l2_norm': 0.0001265075441904225, 'relative_error':
6.940411521624978e-06}]

strict profile iterations: 1

iteration 1: best=(0.2, 0.030154944961) cost=1.49807485555 rotor_range=(0.195002269745, 0.2)
gimbal_range=(0.025156261032, 0.035153628891) expansions=[]

selected rotor lag 0.2 s (40.0182 publish periods)
selected gimbal lag 0.030154944961 s (6.03258 publish periods)
selected objective cost 1.49807475452